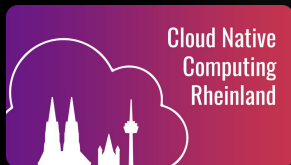




One API to rule them all

Manage Kubernetes Clusters
the Kubernetes way



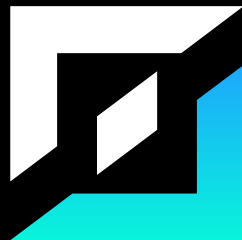
May 12 2025

Tom Wiczorek

Who's that dude?



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Working on
k0S



**CLOUD NATIVE
SANDBOX**



k0SMOTRON

The Problem

- Kubernetes manages workloads well
- But managing Kubernetes clusters requires separate tooling
- This breaks the declarative, GitOps-friendly model
- Leads to duplicated workflows and inconsistent management

What if ...

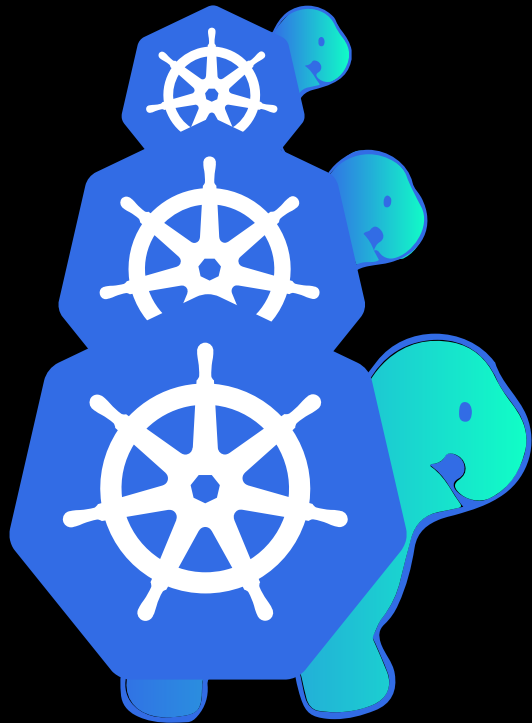


... you could manage clusters just like pods, services, or deployments?

... there was a way to do declarative, API-driven, lifecycle-aware cluster management?

... there was a single model for everything: infrastructure, clusters, workloads?

Turtles all the way down – Say Hi to Cluster API (CAPI)



- Kubernetes subproject
13k+ commits / 500+ contributors
- Declarative clusters management
using Kubernetes-style APIs (CRDs)
- Clusters become just another
Kubernetes resource
- Use Kubernetes to manage
other Kubernetes clusters

Declarative Cluster Management

Before

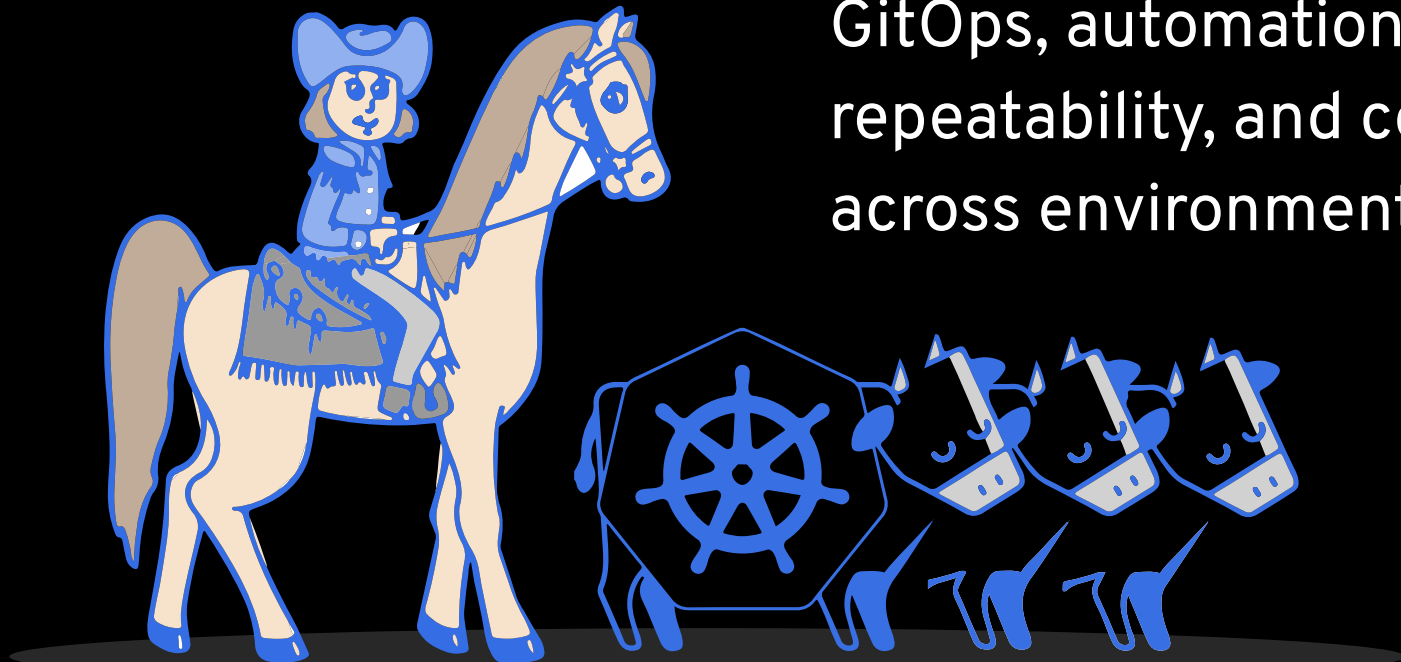
- Manually provision VMs or cloud infrastructure
- Running kubeadm by hand
- Scripts to configure nodes
- Log into machines to upgrade clusters

After

- Define some CRDs in YAML
- Apply them with `kubectl`
- Let CAPI reconcile them
- Manage upgrades and scaling declaratively
- Re-use existing k0s tooling

Declarative Cluster Management – End result

GitOps, automation,
repeatability, and consistency
across environments.

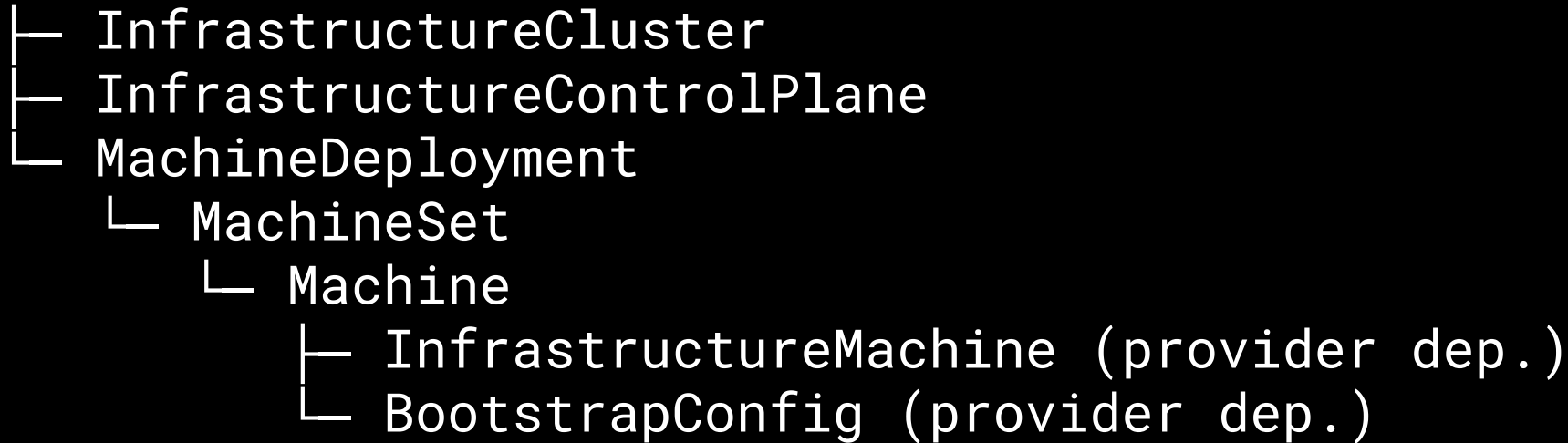


Architecture

- The API of Cluster API (CRDs)
- `clusterctl`
- Pluggable Architecture via Providers
- Isolates infrastructure parts so they can be swapped
- Management clusters vs. workload clusters

CAPI Core CRDs

Cluster



Pluggable Architecture – CAPI Provider Kinds

- Infrastructure

- Machine provisioning
- Bridge to cloud providers / bare metal providers

- Bootstrap

- Turns machines into nodes
- Mostly cloud-init based

- Control plane

- Workload cluster control plane management

- IP address management

- based on IPAddressClaim and IPAddress

- Runtime extensions

- 3rd party extension points

CAPI Cluster Kinds – and

(bootstrap)

Initially holds all CAPI components
Deploys the mgmt c.
Can be “pivoted” into the mgmt c.
→ Mgmt c. becomes self-contained

management

Has all the CRDs
Runs CAPI Core
Runs CAPI Providers
manages lifecycle of other clusters

N × workload

managed by CAPI
CRDs live in management cluster
run user workloads
don't run any CAPI components

How It Works (Bird's Eye View)

- clusterctl creates provider components
- Controller watches CRDs
- Infra resources (VMs, load balancers, etc.) created via Provider controllers
- Full lifecycle: create, scale, upgrade, delete

Typical ways to install CAPI

- Using `clusterctl`:
 `clusterctl init \`
 `--bootstrap <provider> \`
 `--control-plane <provider> \`
 `--infrastructure <provider>`
- Declaratively using the Cluster API Operator

Separating Clusters and Infrastructure

- Decouple workload from infra configuration
- Reuse cluster definitions across environments
- Enables hybrid and multi-cloud flexibility
- GitOps-ready: Store your clusters in Git

CAPI Acronyms

CAPI → Cluster API core

CAP* → Infrastructure providers

CAB* → Bootstrap providers

CAIPAM* → IP address management providers

CARE* → Runtime extension providers

CAPI 🧡 Acronyms – Infrastructure providers

CAPA	AWS
CAPZ	Azure
CAPG	Google Cloud
CAPDO	DigitalOcean
CAPH	Hetzner
CAPO	OpenStack
CAPV	vSphere
CAPX	Nutanix

CAPVC	vCluster
CAPM3	Metal ³
CAPT	Tinkerbell
CAPK0S	K0smotron
CAPK	KubeVirt
CAPD	Docker
CAPN	Nested Kubernetes
...	

CAPI 💖 Acronyms – Even more

Bootstrap providers

CABPK	kubeadm
CABPM	MicroK8s
CABPKK3S	KubeKey / k3s
CABPK0S	K0smotron

And more ...

CAIPAMIC	IPAM In Cluster
CAIPAMX	IPAM Nutanix
CAREX	Nutanix RT Ext.
KCP	kubeadm control plane provider

👐 on – Demo time

- Everybody talks about ~~At~~ Edge – let's do ~~At~~ Edge
- Deploy a Raspberry Pi single-node cluster via CAPI
- No “cloud” involved
- Uses k0smotrons RemoteMachine concept
- Just scratching the surface of CAP capabilities

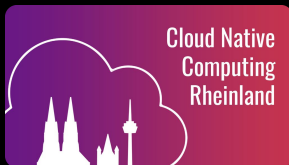
Demo Plan

- Start local management cluster (k0s)
- Install `clusterctl`
- Install CAPI/k0smotron into the management cluster
- Define a single-node cluster using k0smotron
- Provision the Raspberry Pi's SSH key
- See how the cluster gets installed
- Connect to the new workload cluster on the Pi

And that's all there is ... Thanks 🖐️

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Some photos on the slides are from Unsplash – many thanks to the artists!